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Renovation and Sustainable Development Initiative for Mirpur

Introduction:

As we stand at the threshold of the 21st century, the challenges posed by rapid urbanization, environmental degradation, and inadequate infrastructure demand innovative solutions. Nowhere is this more evident than in our own community of Mirpur, specifically in West Monipur, Dhaka-1216. Characterized by haphazard development, lack of environmental considerations, and a dearth of modern amenities, our neighborhood grapples with numerous issues that impact the quality of life of its residents.

To address these pressing concerns head-on, we propose a comprehensive renovation and sustainable development initiative for Mirpur. This initiative aims not only to revitalize our community but also to transform it into a beacon of eco-friendly living and urban excellence. By embracing principles of sustainability, innovation, and community engagement, we envision a future where every resident can thrive in a clean, green, and well-equipped environment.

Objective:

The objective of this project is to transform Mirpur into a model society by implementing the following SMART goals:

Specific: Redesign and rebuild the existing infrastructure of Mirpur to ensure optimal land utilization, efficient transportation systems, and enhanced safety standards.

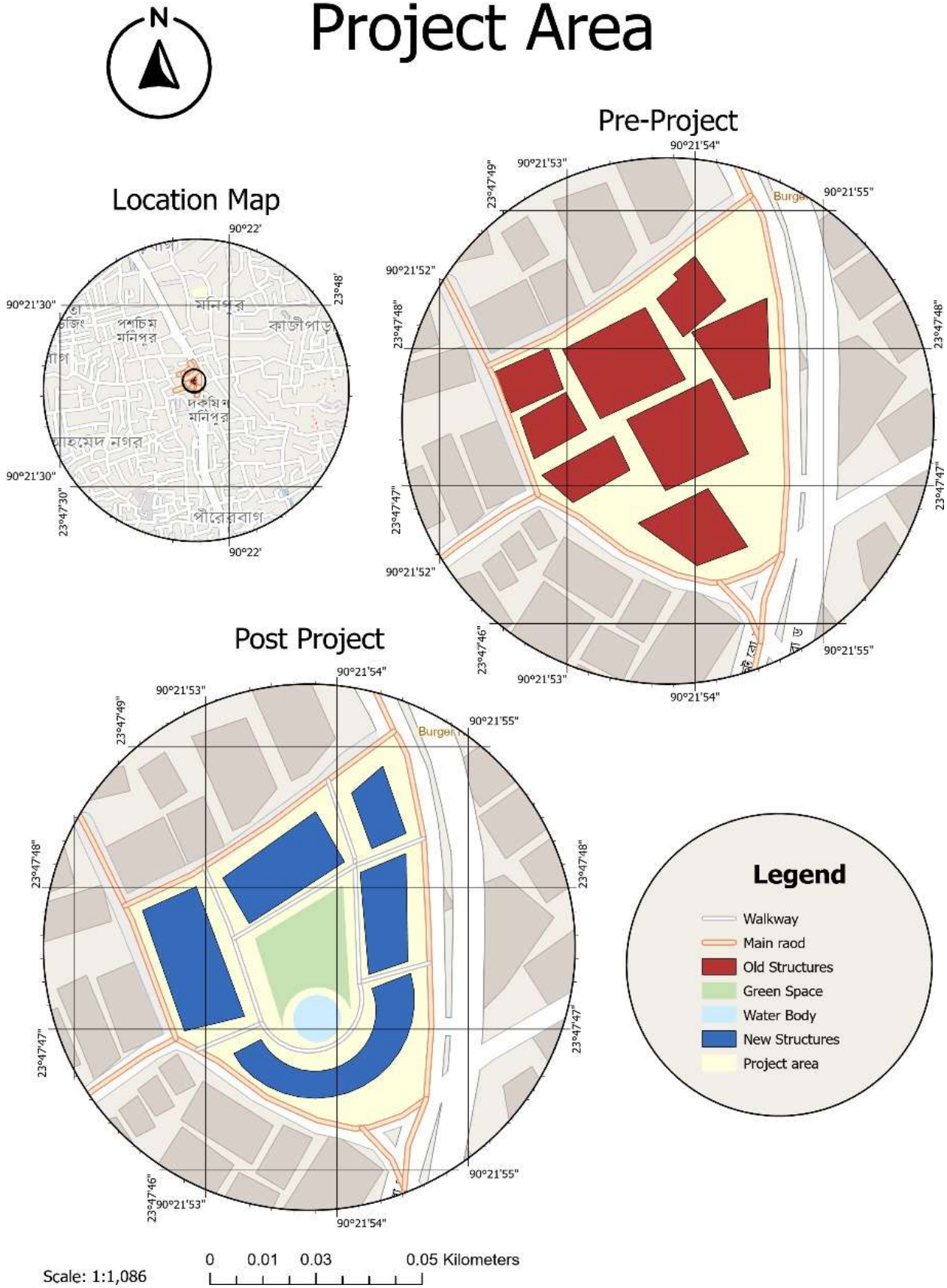
Measurable: Increase green spaces by 30%, reduce air and water pollution levels by 50%, and improve safety ratings by 25% within the first two years of project completion.

Achievable: Collaborate with urban planners, architects, environmental experts, and local authorities to develop feasible blueprints and strategies for the redevelopment process.

Relevant: Address the pressing issues of overpopulation, pollution, and inadequate infrastructure to enhance the quality of life for Mirpur's residents and promote sustainable urban development.

Time-bound: Complete the redevelopment project within a defined timeline of five years, ensuring timely execution of each phase and monitoring progress regularly to meet set milestones.

Project Area



Activities/ Events with logical Sequencing:

Phase I: Pre-Demolition

1. Survey the area: Assess the existing structure and surrounding environment.
2. Inform the Local: Notify authorities and potential residents.
3. Select the method of demolition: Choose the safest and most efficient demolition technique.

Phase II: Demolition

1. Contract Documentation: Prepare legal agreements for demolition services.
2. Tender: Issue a public invitation for qualified demolition contractors to submit bids.
3. Notification of Award: Select and notify the winning demolition contractor.
4. Prepare the site: Secure the area, establish access points, and address safety concerns.
5. Disconnect Utility Supplies: Decommission and safely isolate utilities like gas, water, and electricity.
6. Soft Strip: Remove non-structural elements like windows, doors, and fixtures.
7. Remove Hazardous Materials: Identify, remove, and dispose of asbestos, lead, or other hazardous materials.
8. Final Site Check: Conduct a thorough inspection to ensure the site is ready for demolition.
9. Demolish the site: Execute the chosen demolition method to safely bring down the structure.

Phase III: Post-Demolition

1. Material Sorting and Documentation: Categorize salvageable materials and debris for recycling or disposal.
2. Tender: Invite bids for waste removal and potentially, material resale.
3. Notification of Award: Select and notify the winning contractor for debris removal and material sales.
4. Clear the site: Remove all debris and prepare the ground for new construction.

Phase IV: Redevelopment

1. Design the Residential Society: Develop plans for the new residential development.
2. Detailing & Contract Documentation: Define project specifications and prepare construction contracts.
3. Tender: Issue a public invitation for qualified construction companies to bid on the project.
4. Notification of Award: Select and notify the winning construction contractor.
5. Construction of New Residential Society: Build the new residential development according to the approved design.

Cost Estimation and Budgeting:

Phase I: Pre-Demolition (5% of Total Budget)

Category	Estimated Amount	Estimated Percentage	Description
Labor	600,000.00৳	20%	Survey crew, notification preparation, demolition planning team
Materials	300,000.00৳	10%	Signage, fencing, permits
Equipment	300,000.00৳	10%	Survey equipment, communication tools
Other	1,800,000.00৳	60%	Environmental assessments, asbestos testing (if applicable)

Phase II: Demolition (40% of Total Budget)

Category	Estimated Amount	Estimated Percentage	Description
Labor	9,600,000.00৳	40%	Demolition crew, safety personnel, environmental monitors (if applicable)
Materials	4,800,000.00৳	20%	Consumables (fuel, filters), demolition tools, personal protective equipment (PPE)
Equipment	6,000,000.00৳	25%	Demolition machinery (excavators, loaders), waste transportation vehicles
Permits & Licenses	3,600,000.00৳	15%	Demolition permits, waste disposal permits

Phase III: Post-Demolition (10% of Total Budget)

Category	Estimated Amount	Estimated Percentage	Description
Labor	1,800,000.00৳	30%	Sorting crew, waste removal team
Materials	600,000.00৳	10%	Waste disposal fees, recycling fees
Equipment	2,400,000.00৳	40%	Waste hauling trucks, sorting equipment (if applicable)
Landfill Fees	1,200,000.00৳	20%	Depending on the amount and type of debris

Phase IV: Redevelopment (45% of Total Budget)

Category	Estimated Amount	Estimated Percentage	Description
Design & Engineering	2,700,000.00৳	10%	Architectural and engineering fees
Materials	16,200,000.00৳	60%	Building materials, fixtures, appliances
Labor	5,400,000.00৳	20%	Construction crew, subcontractors
Permits & Inspections	2,700,000.00৳	10%	Building permits, construction inspections

List of Required Personnel:

Phase I: Pre-Demolition (Estimated Duration: 2-4 Weeks)

Position	Skills	Service Duration (Weeks)
Project Manager	Construction management, demolition planning, permitting experience	4
Surveyor	Land surveying, data analysis	2
Environmental Consultant (if applicable)	Asbestos testing, lead paint inspection	1-2 (depending on scope)

Phase II: Demolition (Estimated Duration: 4-8 Weeks)

Position	Skills	Service Duration (Weeks)
Demolition Supervisor	Demolition experience, safety protocols, crew management	8
Demolition Crew	Heavy machinery operation, demolition techniques, safety training	4-6
Safety Officer	Construction safety regulations, hazard identification, first aid	8
Environmental Monitor (if applicable)	Asbestos abatement monitoring, lead dust monitoring	2-4 (depending on scope)

Phase III: Post-Demolition (Estimated Duration: 2-4 Weeks)

Position	Skills	Service Duration (Weeks)
Site Supervisor	Waste management experience, debris sorting protocols	4
Sorting Crew	Manual labor, material identification, PPE usage	2-4
Waste Hauler	Commercial driver's license, experience with waste disposal regulations	2 (on-call basis)

Phase IV: Redevelopment (Estimated Duration: 36-52 Weeks)

Position	Skills	Service Duration (Weeks)
Architect	Architectural design, building code compliance	12-16
Structural Engineer	Structural engineering expertise, foundation design	8-12
Civil Engineer	Site planning, utilities coordination	4-8
General Contractor	Construction management, subcontractor coordination	52
Construction Crew (Variable)	Carpentry, electrical, plumbing, HVAC, etc.	36-52 (depending on specific trades)
Inspectors	Building code enforcement, safety inspections	Variable (periodic visits)

List of Equipment, Materials, and Technologies

Phase I: Pre-Demolition

Category	Equipment/ Materials/ Technologies	Description
Equipment	Surveying equipment (total stations, GPS)	Site measurement and data collection
	Communication tools (radios, cell phones)	Project communication and coordination
Materials	Signage (project information, safety warnings)	Public notification and safety demarcation
	Fencing (perimeter control)	Securing the demolition site

Phase II: Demolition

Category	Equipment/ Materials/ Technologies	Description
Equipment	Demolition machinery (excavators, loaders, hydraulic hammers, TNTs)	Safe and efficient structure disassembly
	Safety equipment (fall protection, dust masks, hard hats)	Worker safety during demolition activities
Materials	Dust suppression systems (water cannons, misting units)	Mitigating airborne dust particles
	Cutting tools (diamond saws, torches)	Disconnecting utilities and selective demolition
	Consumables (fuel, filters, lubricants)	Maintaining operational efficiency of machinery

Phase III: Post-Demolition

Category	Equipment/ Materials/ Technologies	Description
Equipment	Waste sorting equipment (conveyors, sorting bins)	Separating recyclable materials from debris
	Safety equipment (fall protection, dust masks, hard hats)	Worker safety during demolition activities
	Waste hauling trucks	Transporting debris to landfills or recycling facilities
Materials	Personal protective equipment (PPE)	Worker safety during sorting and waste handling
	Liners for waste transportation	Containing and securing debris during transport
	Recycling bins and signage	Facilitating proper waste segregation for recycling

Phase IV: Redevelopment

Category	Equipment/ Materials/ Technologies	Description
Technologies	Building Information Modeling (BIM)	3D modelling software for design coordination and clash detection
	Project management software	Streamlining communication, scheduling, and resource allocation
Equipment	Construction machinery (cranes, bulldozers, concrete mixers)	Building construction activities
	Power tools (drills, saws, nail guns)	Installation and assembly tasks
	Surveying equipment (for verification and quality control)	Ensuring construction adheres to design plans
Materials	Building materials (concrete, steel, lumber, etc.)	Components for the new residential development
	Interior finishes (flooring, paint, fixtures)	Completing the new residential units

Network Diagram:

Activity Oriented

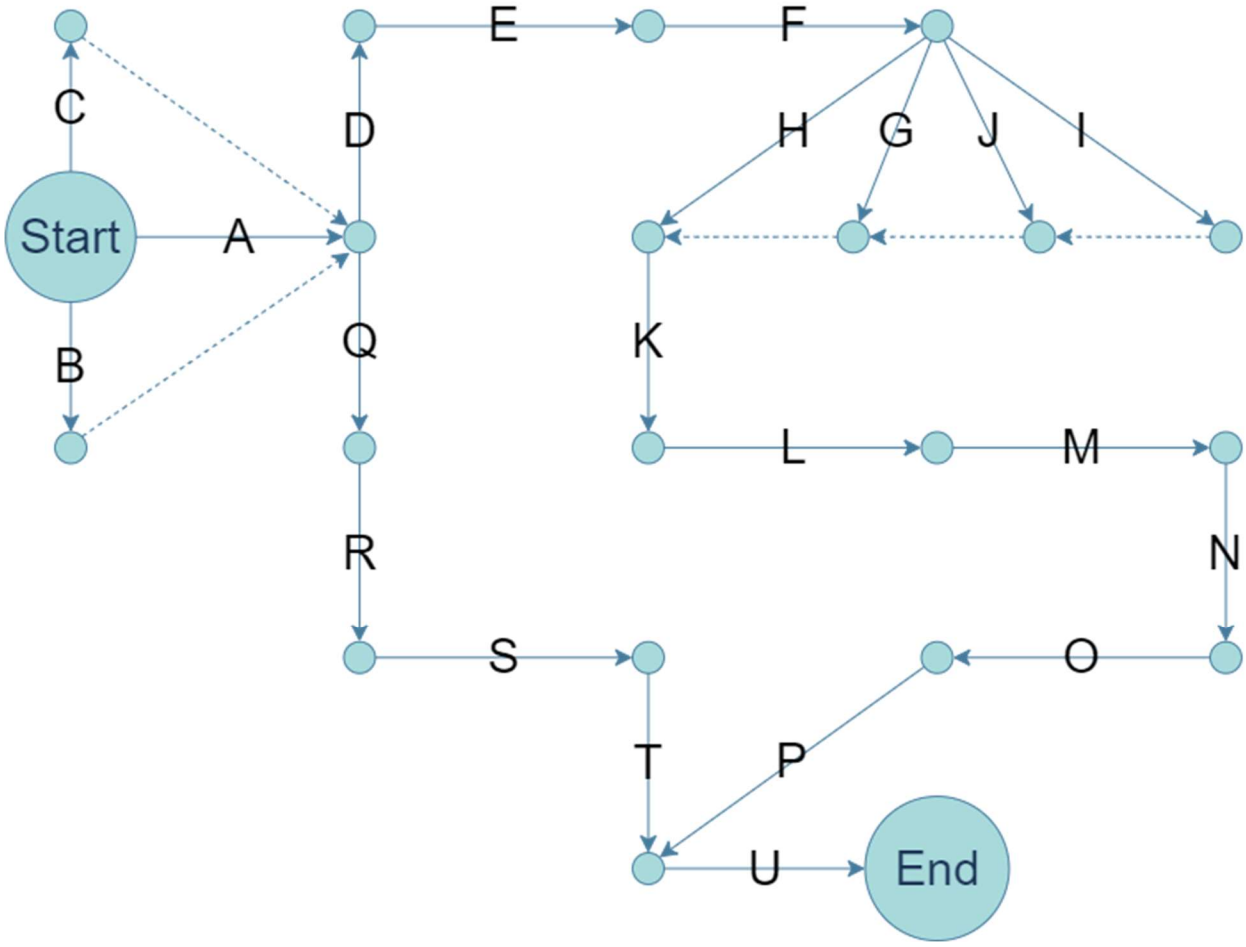


Figure 1: Activity Oriented Network Diagram

Activity	Details	Phase
A	Survey the area	Phase I
B	Inform the Local	Phase I
C	Select the method of demolition	Phase I
D	Contract documentation for demolition	Phase II
E	Tender for demolition	Phase II
F	NOA for demolition	Phase II
G	Prepare the site	Phase II
H	Disconnect utility supplies	Phase II
I	Soft strip	Phase II
J	Remove hazardous material	Phase II
K	Final site check	Phase II
L	Demolish the site	Phase II
M	Material sorting and post demolition documentation	Phase III
N	Tender for post demolition work	Phase III
O	NOA for post demolition work	Phase III
P	Clear the site for redevelopment	Phase III
Q	Design the residential society	Phase IV
R	Detailing & contract documentation for redevelopment	Phase IV
S	Tender for redevelopment	Phase IV
T	NOA for redevelopment	Phase IV
U	Construction of new residential society	Phase IV

Chart 2: Activity Details

In Figure 1, it shows the activity-oriented diagram and the chart 1 shows the activity symbol and details of activity.

Event Oriented

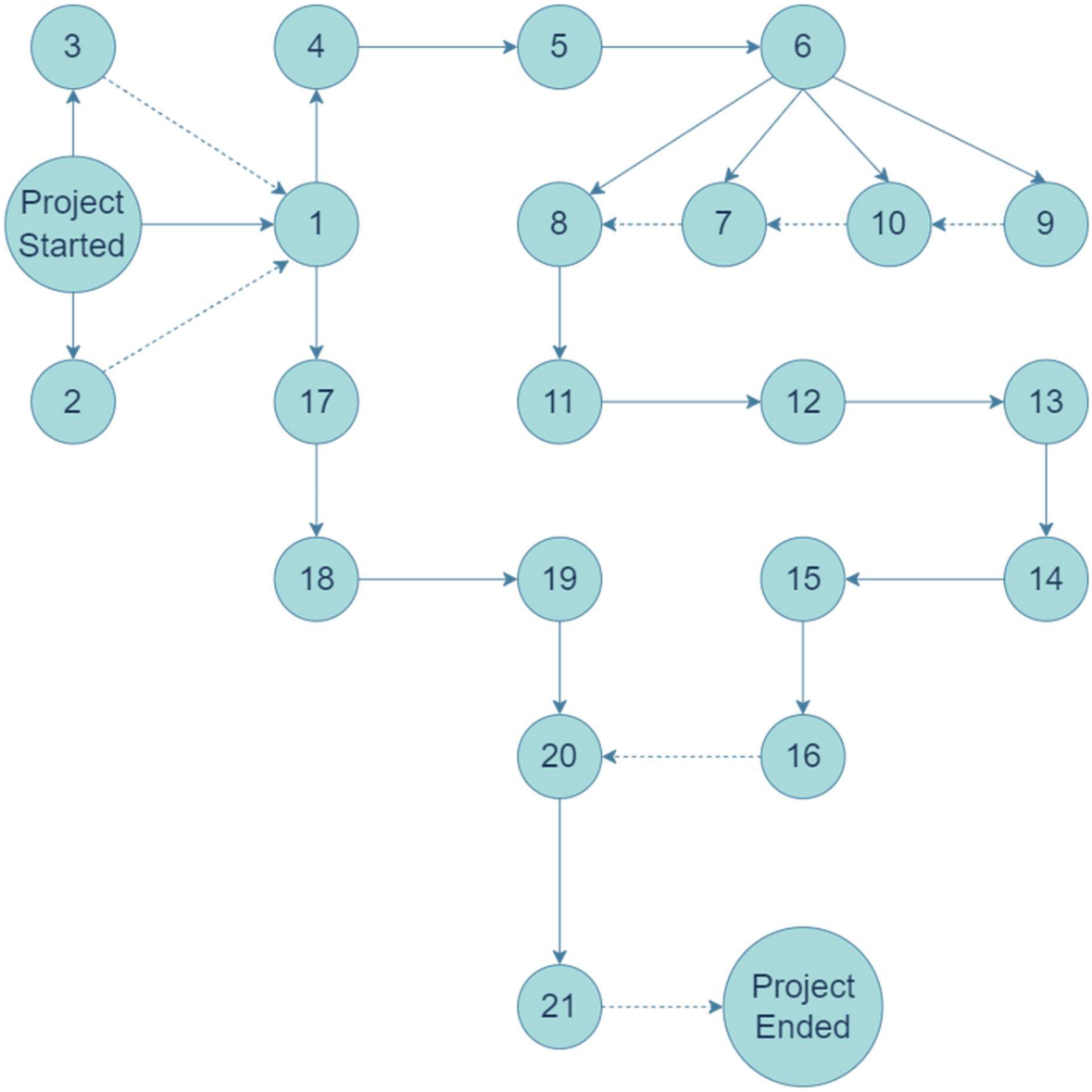


Figure 3: Event Oriented Network Diagram

Event	Details	Phase
1	Area surveyed	Phase I
2	Locals informed	Phase I
3	Demolition method selected	Phase I
4	Contract documentation completed	Phase II
5	Demolition tendered	Phase II
6	NOA of demolition sent	Phase II
7	Site prepared	Phase II
8	Utility supplies disconnected	Phase II
9	Soft stripped	Phase II
10	Hazardous materials removed	Phase II
11	Site check completed	Phase II
12	Demolished	Phase II
13	Material sorting and post demolition documentation completed	Phase III
14	Tender opened for post demolition work	Phase III
15	NOA of post demolition work sent	Phase III
16	Site cleared	Phase III
17	Residential society designed	Phase IV
18	Detailing & contract documentation completed	Phase IV
19	Redevelopment work tendered	Phase IV
20	NOA of redevelopment work sent	Phase IV
21	New residential society constructed	Phase IV

Chart 2: Activity Details

In Figure 2, it shows the event-oriented diagram and the chart 2 shows the activity symbol and details of events.

Time Estimation for Activities:

Phase I: Pre-Demolition

Activity	Optimistic Time (in weeks)	Most Likely Time (in weeks)	Pessimistic Time (in weeks)
Survey the area	1	2	3
Inform the Local	1	2	4
Select the method of demolition	1	2	4
Total	3	6	11

Phase II: Demolition

Activity	Optimistic Time (in weeks)	Most Likely Time (in weeks)	Pessimistic Time (in weeks)
Contract Documentation	1	2	3
Tender	1	2	4
Notification of Award	1	2	4
Prepare the site	1	2	4
Disconnect Utility Supplies	1	2	4
Soft Strip (windows, doors, fixtures)	2	4	6
Remove Hazardous Materials (if applicable)	2	8	12
Final Site Check	1	2	3
Demolish the site	2	4	8

Phase III: Post-Demolition

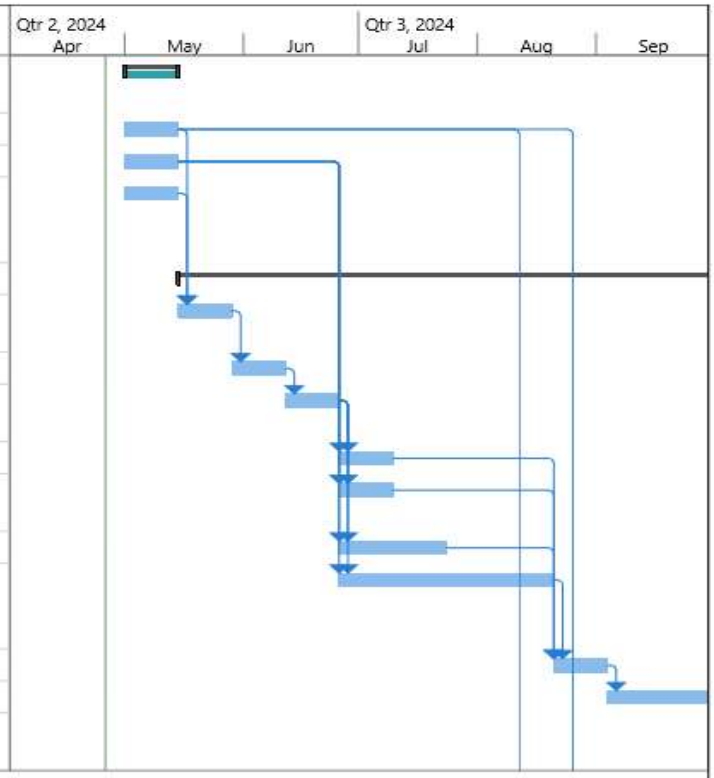
Activity	Optimistic Time (in weeks)	Most Likely Time (in weeks)	Pessimistic Time (in weeks)
Detailing and contract documentation of materials	1	2	4
Tender	1	2	4
Notification of Award	1	2	4
Clear the Site	1	2	4

Phase IV: Redevelopment

Activity	Optimistic Time (in weeks)	Most Likely Time (in weeks)	Pessimistic Time (in weeks)
Design the Residential Society	4	8	12
Detailing & Contract Documentation	4	8	12
Tender	2	4	6
Notification of Award	1	2	4
Construction of New Residential Society	36	48	60

Gantt Chart:

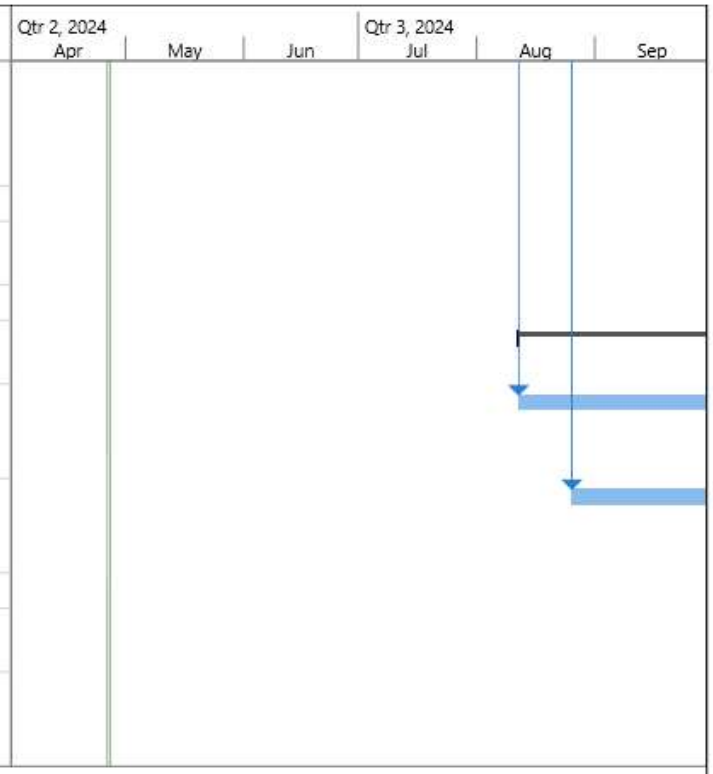
ID		Task Mode	Task Name	Duration	Start	Finish	Predecessors
1			**Phase I: Pre-Demolition**	**10 days**	**Wed 5/1/24**	**Tue 5/14/24**	
2			Survey the area	2 wks	Wed 5/1/24	Tue 5/14/24	
3			Inform the local	2 wks	Wed 5/1/24	Tue 5/14/24	
4			Select the method of demolition	2 wks	Wed 5/1/24	Tue 5/14/24	
5			**Phase II: Demolition**	**100 days**	**Wed 5/15/24**	**Tue 10/1/24**	
6			Contract Documentation	2 wks	Wed 5/15/24	Tue 5/28/24	4,2
7			Tender	2 wks	Wed 5/29/24	Tue 6/11/24	6
8			Notification of Award	2 wks	Wed 6/12/24	Tue 6/25/24	7
9			Prepare the site	2 wks	Wed 6/26/24	Tue 7/9/24	8,3
10			Disconnect Utiliy Suppies	2 wks	Wed 6/26/24	Tue 7/9/24	8,3
11			Soft Stirp	4 wks	Wed 6/26/24	Tue 7/23/24	8,3
12			Remove Hazardouse Materials	8 wks	Wed 6/26/24	Tue 8/20/24	8,3
13			Final Site Check	2 wks	Wed 8/21/24	Tue 9/3/24	9,10,11,12
14			Demolish the site	4 wks	Wed 9/4/24	Tue 10/1/24	13
15			**Phase III: Post-Demolition**	**40 days**	**Wed 10/2/24**	**Tue 11/26/24**	



Project: Project1
Date: Fri 4/26/24

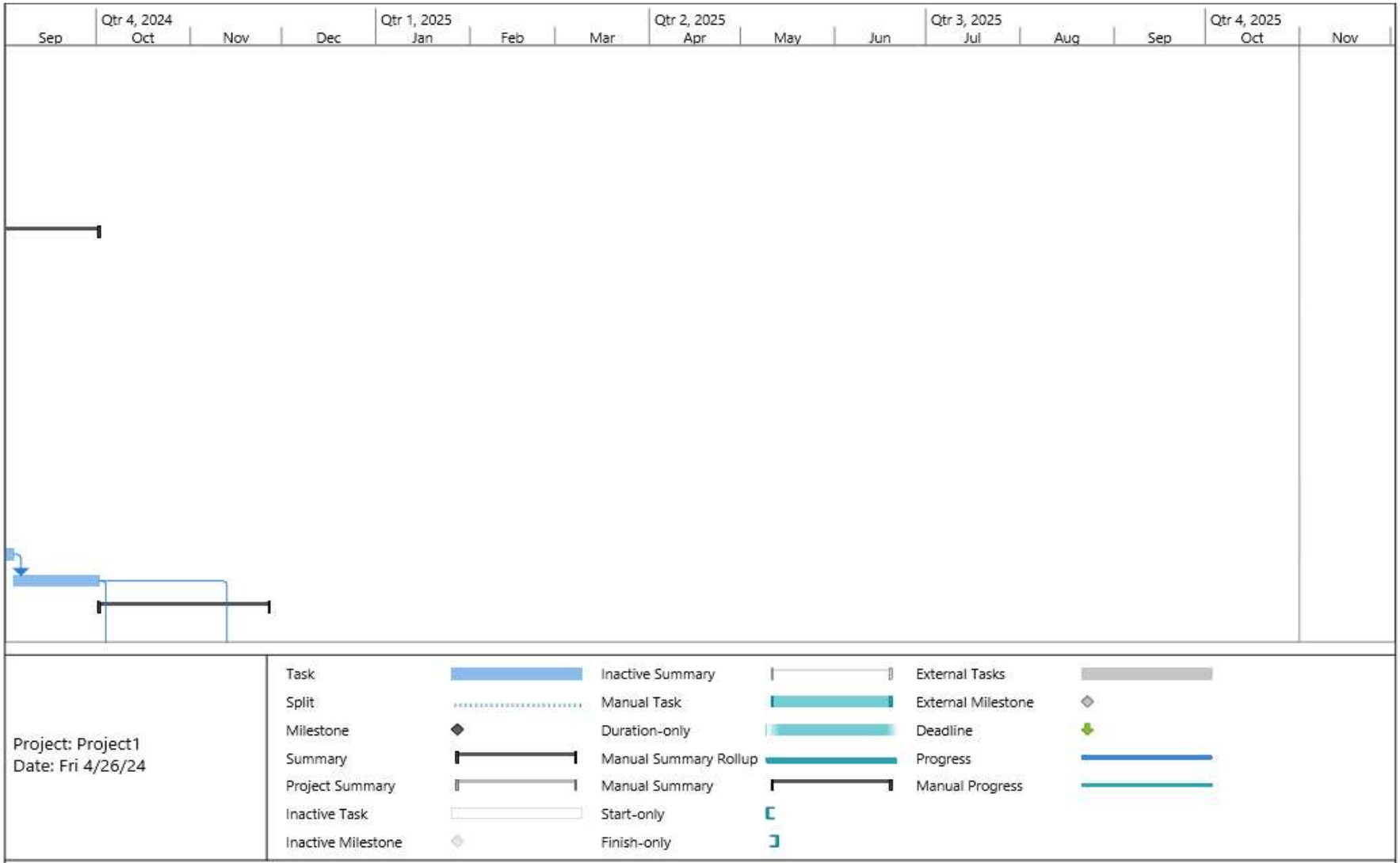
Task	Inactive Summary	External Tasks	
Split	Manual Task	External Milestone	
Milestone	Duration-only	Deadline	
Summary	Manual Summary Rollup	Progress	
Project Summary	Manual Summary	Manual Progress	
Inactive Task	Start-only		
Inactive Milestone	Finish-only		

ID		Task Mode	Task Name	Duration	Start	Finish	Predecessors
16			Detailing and contract documenttion of materials	2 wks	Wed 10/2/24	Tue 10/15/24	14
17			Tender	2 wks	Wed 10/16/2	Tue 10/29/24	16
18			Notification of Award	2 wks	Wed 10/30/24	Tue 11/12/24	17
19			Clear the site	2 wks	Wed 11/13/2	Tue 11/26/24	14,18
20			**Phase IV: Redevelopment**	**320 days**	**Mon 8/12/24**	**Fri 10/31/25**	
21			Design the Residential Society	8 wks	Mon 8/12/24	Fri 10/4/24	2
22			Detailing & Contract Documentation	8 wks	Mon 8/26/24	Fri 10/18/24	2
23			Tender	4 wks	Mon 10/21/2	Fri 11/15/24	21,22
24			Notification of Award	2 wks	Mon 11/18/24	Fri 11/29/24	23
25			Construction of New Residential society	48 wks	Mon 12/2/24	Fri 10/31/25	24,19



Project: Project1
Date: Sat 4/27/24

Task		Inactive Summary		External Tasks	
Split		Manual Task		External Milestone	
Milestone		Duration-only		Deadline	
Summary		Manual Summary Rollup		Progress	
Project Summary		Manual Summary		Manual Progress	
Inactive Task		Start-only			
Inactive Milestone		Finish-only			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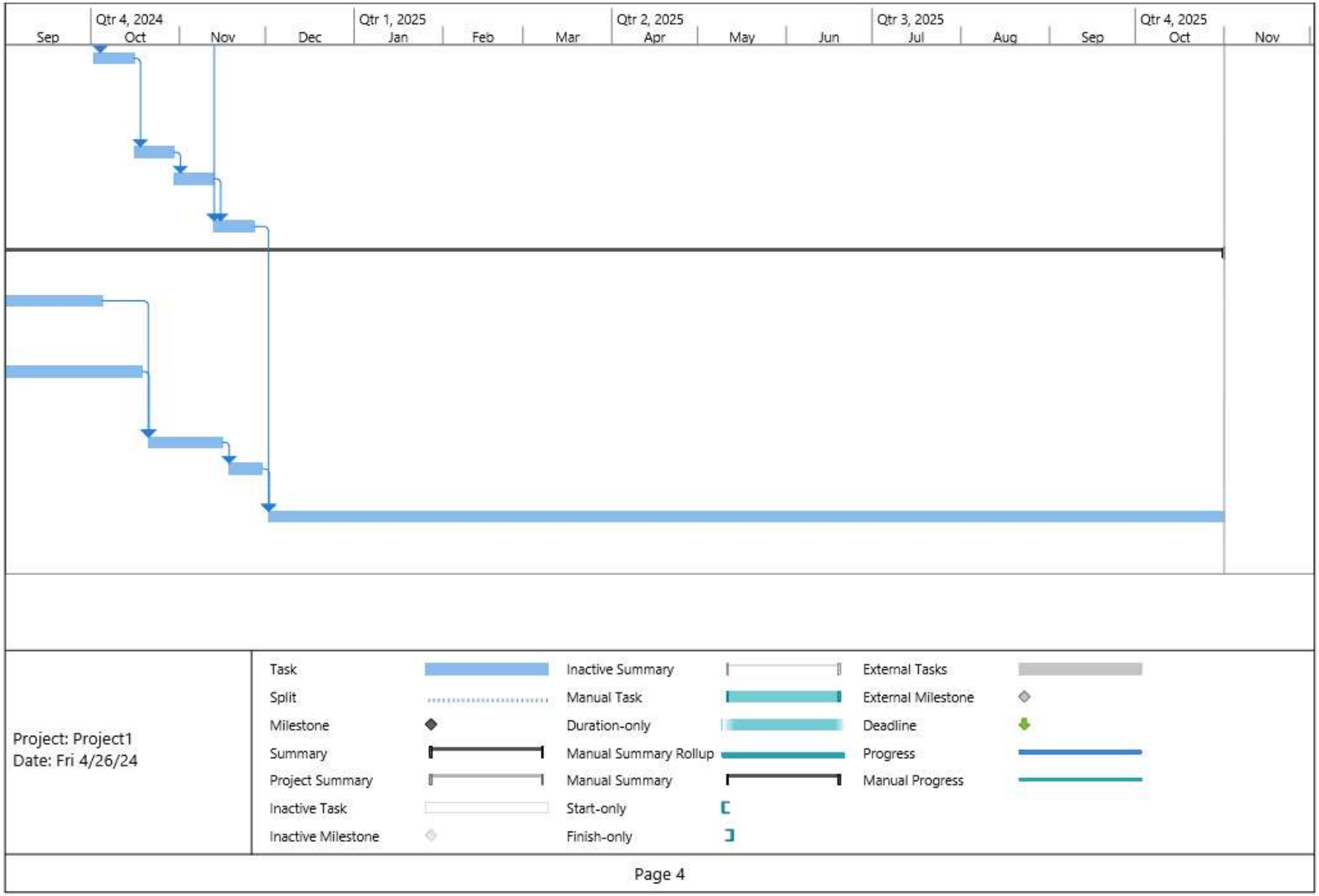
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Duration-only
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Manual Summary
Start-only
Finish-only
External Tasks
External Milestone
Deadline
Progress
Manual Progress
Page 4

Expected time per activity:

Activity	Details	Phase	t_0	t_l	t_p	t_e(wks)
A	Survey the area	Phase I	1	2	3	2.00
B	Inform the Local	Phase I	1	2	4	2.17
C	Select the method of demolition	Phase I	1	2	4	2.17
D	Contract documentation for demolition	Phase II	1	2	3	2.00
E	Tender for demolition	Phase II	1	2	4	2.17
F	NOA for demolition	Phase II	1	2	4	2.17
G	Prepare the site	Phase II	1	2	4	2.17
H	Disconnect utility supplies	Phase II	1	2	4	2.17
I	Soft strip	Phase II	2	4	6	4.00
J	Remove hazardous material	Phase II	2	8	12	7.67
K	Final site check	Phase II	1	2	3	2.00
L	Demolish the site	Phase II	2	4	8	4.33
M	Material sorting and post demolition documentation	Phase III	1	2	4	2.17
N	Tender for post demolition work	Phase III	1	2	4	2.17
O	NOA for post demolition work	Phase III	1	2	4	2.17
P	Clear the site for redevelopment	Phase III	1	2	4	2.17
Q	Design the residential society	Phase IV	4	8	12	8.00
R	Detailing & contract documentation for redevelopment	Phase IV	4	8	12	8.00
S	Tender for redevelopment	Phase IV	2	4	6	4.00
T	NOA for redevelopment	Phase IV	1	2	4	2.17
U	Construction of new residential society	Phase IV	36	48	60	48.00

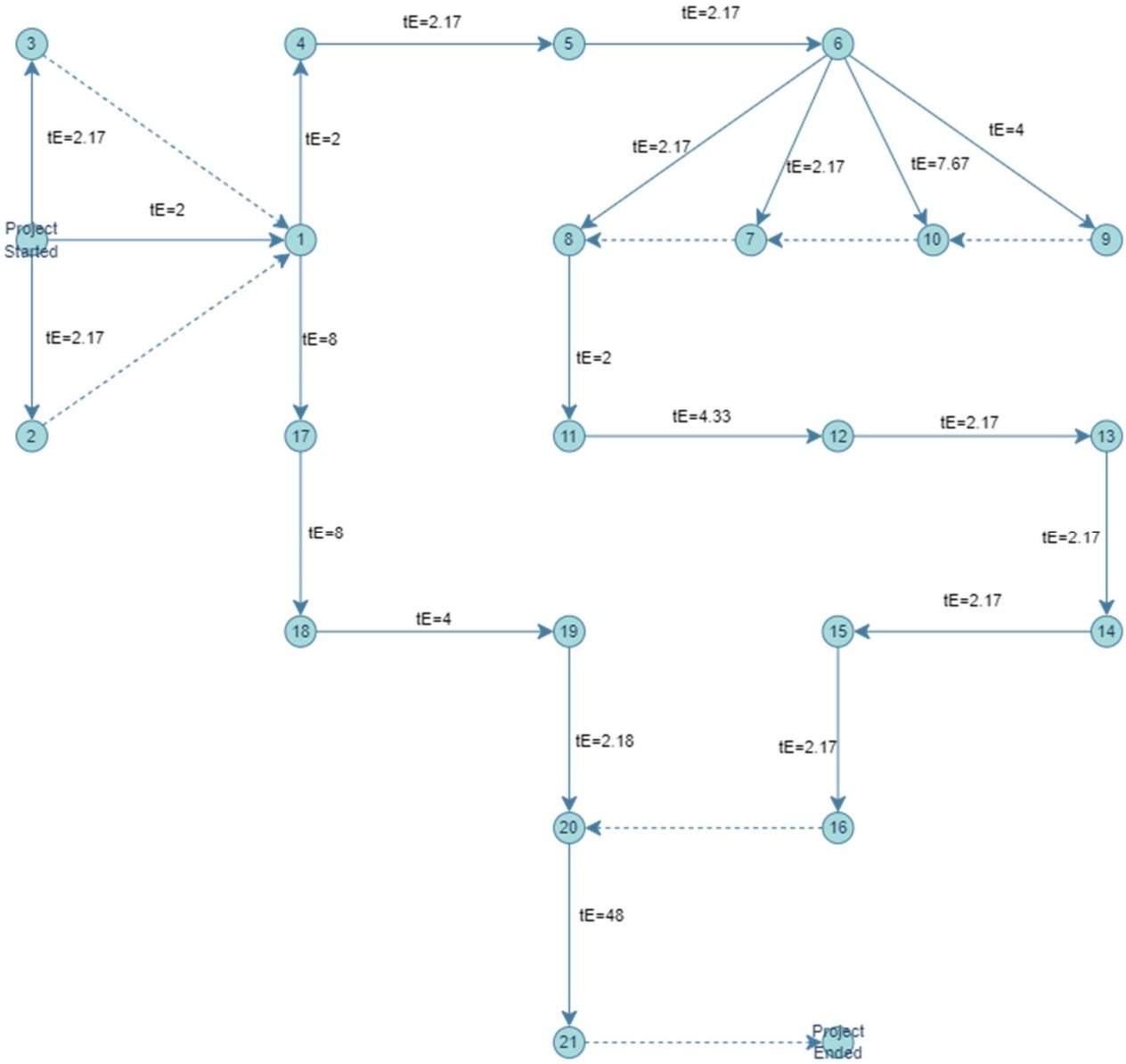

Figure 3: Diagram for Critical Path Calculation

Figure 3 Show the expected time of every critical path. The critical path is the longest path through the network and time along this path gives the project duration.

Critical Path:

Path	Expected Time (t_E)
I (Start-1-4~6-8~16-20~End)	73.52
I2 (Start-2-1-4~6-8~16-20~End)	73.69
I3 (Start-3-1-4~6-8~16-20~End)	73.69
II (Start-1-4~6-7-8~16-20~End)	73.52
II2 (Start-1-4~6-10-7-8~16-20~End)	79.02
II2 (Start-2or3-4~6-10-7-8~16-20~End)	79.19
III (Start-1-4~6-9-10-7-8~16-20~End)	75.35
III2 (Start-2or3-4~6-9-10-7-8~16-20~End)	75.52
IV (Start-1-17~20~End)	72.17
IV2 (Start-2or3-17~20~End)	72.34

From the above table, we find that path II2 is the critical since t_E is the maximum.

Earliest Expected Time and the Latest Allowable Occurrence Time (from network diagram):

Event	i	Activity	t_E	T_E	j	Activity	t_E	TL	Slack
Start	-		0	0	1	Start-1	2	0.17	
					2	Start-2	2.17	0	
					3	Start-3	2.17	0	0
1	Start	Start-1	2	2	4	1-4	2	2.17	0.17
	2	2-1	0	0	17	1-17	8	9.02	
	3	3-1	0	0					
2	Start	Start-2	2.17	2.17	1	2-1	0	2.17	0
3	Start	Start-3	2.17	2.17	1	3-1	0	2.17	0
4	1	1-4	2	4	5	4-5	2.17	4.17	0
	2	2-4	2	4.17					
	3	3-4	2	4.17					
5	4	4-5	2.17	6.34	6	5-6	2.17	6.34	0
6	5	5-6	2.17	8.51	7	6-7	2.17	14.01	0
					8	6-8	2.17	14.01	
					9	6-9	4	12.18	
					10	6-10	7.67	8.51	
7	6	6-7	2.17	10.68	8	7-8	0	16.18	5.5
8	6	6-8	2.17	10.68	11	8-11	2	16.18	5.5
9	6	6-9	4	12.51	10	9-10	0	16.18	3.76
10	6	6-10	7.67	16.18	7	10-7	0	16.18	0
11	8	8-11	2	12.68	12	11-12	4.33	18.18	0
	7	7-11	2	12.68					
	10	10-11	2	18.18					
	9	9-11	2	14.51					
12	11	11-12	4.33	22.51	13	12-13	2.17	22.51	0
13	12	12-13	2.17	24.68	14	13-14	2.17	24.68	0

14	13	13-14	2.17	26.85	15	14-15	2.17	26.85	0
15	14	14-15	2.17	29.02	16	15-16	2.17	29.02	0
16	15	15-16	2.17	31.19	20	16-20	0	31.19	0
17	1	1-17	8	10	18	17-18	8	17.02	6.85
	2	2-17	8	10.17					
	3	3-17	8	10.17					
18	17	17-18	8	18.17	19	18-19	4	25.02	6.85
19	18	18-19	4	22.17	20	19-20	2.17	29.02	6.85
20	19	19-20	2.17	24.34	21	20-21	48	31.19	0
	16	16-20	0	31.19					
21	20	20-21	48	79.19	End	21-End	0	79.19	0
End	21	21-End	0	79.19				79.19	0

From the above Table, we find that the minimum value of slack is 0 week, and this occurs at events Start-2/3-4-5-6-10-11-12-13-14-15-16-20-21 and End. Hence critical path is along Start-2/3-4-5-6-10-11-12-13-14-15-16-20-21 and End.

The next minimum slack value is 0.17 which occurs at event 1. Hence the path connecting start-1-4… is the sub-critical or semi-critical path.

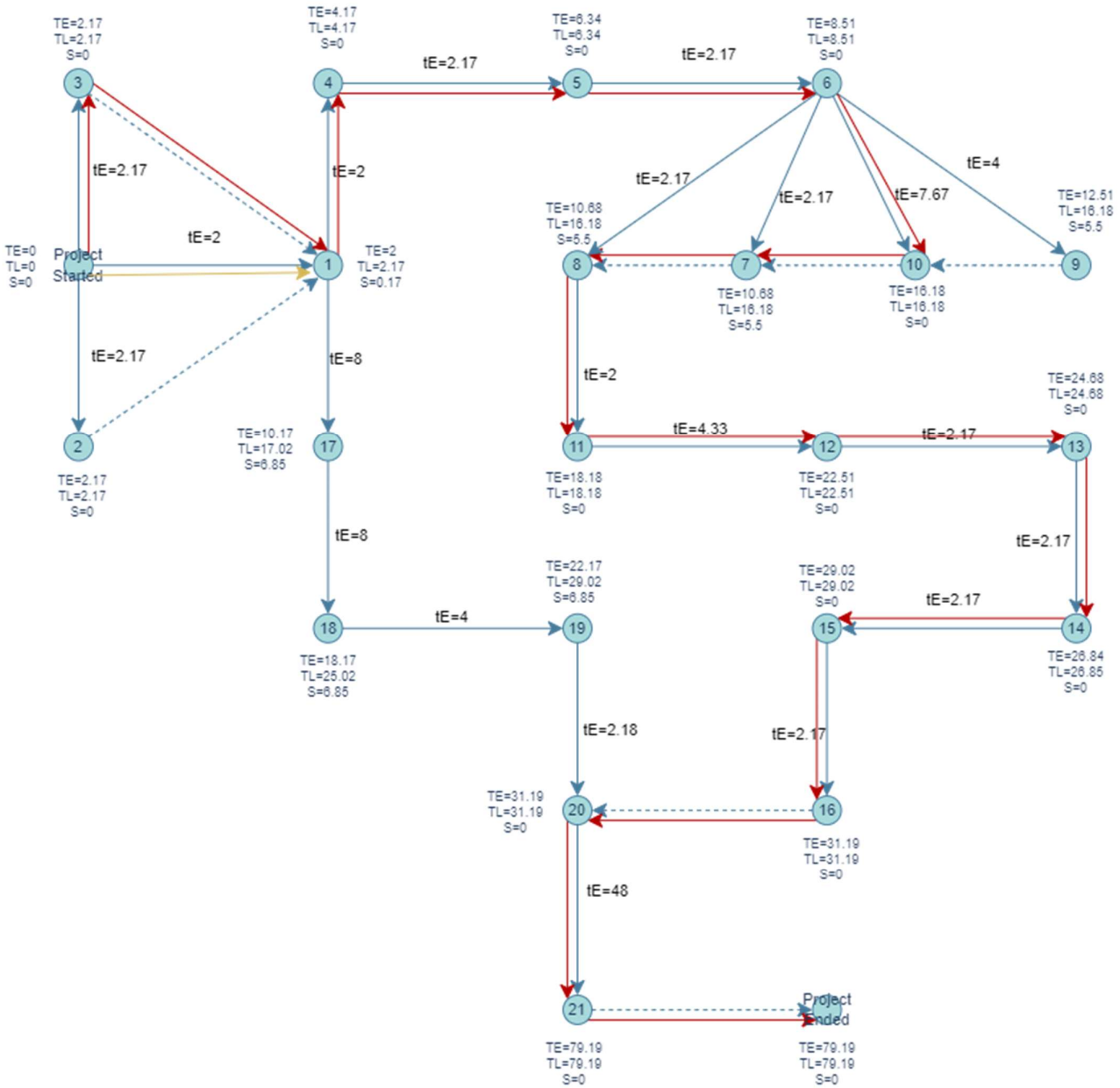

Figure 4: Critical path and Subcritical Path

Figure 4 Show the critical path which is highlighted with red arrow and subcritical path is highlighted with yellow arrow.

Probability of the project's completion time

Here,

Expected time, $T_E = 79.19 \approx 80\ wks$

Standard Deviation $\sigma = \sqrt{\sum \left(\frac{t_p^{ij} - t_o^{ij}}{6}\right)^2}$

Activity		Optimistic Time (in weeks)	Pessimistic Time (in weeks)	$\left(\frac{t_p^{ij} - t_o^{ij}}{6}\right)^2$
Survey the area	A	1	3	0.111111
Inform the Local	B	1	4	0.25
Select the method of demolition	C	1	4	0.25
Contract Documentation	D	1	3	0.111111
Tender	E	1	4	0.25
Notification of Award	F	1	4	0.25
Prepare the site	G	1	4	0.25
Disconnect Utility Supplies	H	1	4	0.25
Soft Strip (windows, doors, fixtures)	I	2	6	0.444444
Remove Hazardous Materials (if applicable)	J	2	12	2.777778
Final Site Check	K	1	3	0.111111
Demolish the site	L	2	8	1

Detailing and contract documentation of materials	M	1	4	0.25
Tender	N	1	4	0.25
Notification of Award	O	1	4	0.25
Clear the Site	P	1	4	0.25
Design the Residential Society	Q	4	12	1.777778
Detailing & Contract Documentation	R	4	12	1.777778
Tender	S	2	6	0.444444
Notification of Award	T	1	4	0.25
Construction of New Residential Society	U	36	60	16
			σ^2	27.30556

Standard Deviation $\sigma = \sqrt{\sum\left(\frac{t_p^{ij}-t_o^{ij}}{6}\right)^2} = \sqrt{27.30556}$ =5.225472

For 95% probability, $Z \approx 1.65$

So,

$T_S = \sigma Z + T_E$
$T_S = 5.33 * 1.65 + 80 = 89\ wks$

For 75% probability, $Z \approx 0.69$

So,

$T_s = \sigma Z + T_E$
$T_s = 5.33 * 0.69 + 80 = 84\ wks$

For 50% probability, $Z \approx 0$

So,

$T_s = \sigma Z + T_E$
$T_s = 5.33 * 0 + 80 = 80\ wks$

Activity times and floats for each activity:

			Earliest		Latest		Float			
Activity (i-j)		Duration (tij)	Start time EST (TEi)	Finish time EFT	Start time LST	Finish time LFT	Total FT	Free FF	Slack i	Independent FID
Start-1	A	2	0	2	0.17	2.17	0.17	0	0	0
Start-2	B	2.17	0	2.17	0	2.17	0	0.17	0	0.17
Start-3	C	2.17	0	2.17	0	2.17	0	0.17	0	0.17
1-4	D	2	2	4	2.17	4.17	0.17	0.17	0.17	0
4-5	E	2.17	4.17	6.34	4.17	6.34	0	0	0	0
5-6	F	2.17	6.34	8.51	6.34	8.51	0	0	0	0
6-7	G	2.17	8.51	10.68	14.01	16.18	5.5	0	0	0
6-8	H	2.17	8.51	10.68	14.01	16.18	5.5	0	0	0
6-9	I	4	8.51	12.51	12.18	16.18	3.67	0	0	0
6-10	J	7.67	8.51	16.18	8.51	16.18	0	0	0	0
9-10	J'	0	12.51	12.51	16.18	16.18	3.67	3.67	3.67	0
10-7	J"	0	16.18	16.18	16.18	16.18	0	-5.5	0	-5.5
7-8	J'"	0	10.68	10.68	16.18	16.18	5.5	0	5.5	-5.5
8-11	K	2	10.68	12.68	16.18	18.18	5.5	5.5	5.5	0
11-12	L	4.33	18.18	22.51	18.18	22.51	0	0	0	0
12-13	M	2.17	22.51	24.68	22.51	24.68	0	0	0	0
13-14	N	2.17	24.68	26.85	24.68	26.85	0	0	0	0
14-15	O	2.17	26.85	29.02	26.85	29.02	0	0	0	0
15-16	P	2.17	29.02	31.19	29.02	31.19	0	0	0	0
1-17	Q	8	2	10	9.02	17.02	7.02	0.17	0.17	0
17-18	R	8	10.17	18.17	17.02	25.02	6.85	0	6.85	-6.85
18-19	S	4	18.17	22.17	25.02	29.02	6.85	0	6.85	-6.85
19-20	T	2.17	22.17	24.34	29.02	31.19	6.85	0	6.85	-6.85
20-21	U	48	31.19	79.19	31.19	79.19	0	0	0	0

In activity times and floats super critical activity has negative float. those activities are occurring later than the scheduled time. Free Float denotes the amount of time an activity's early start can be delayed without affecting the early start of its successor activity.

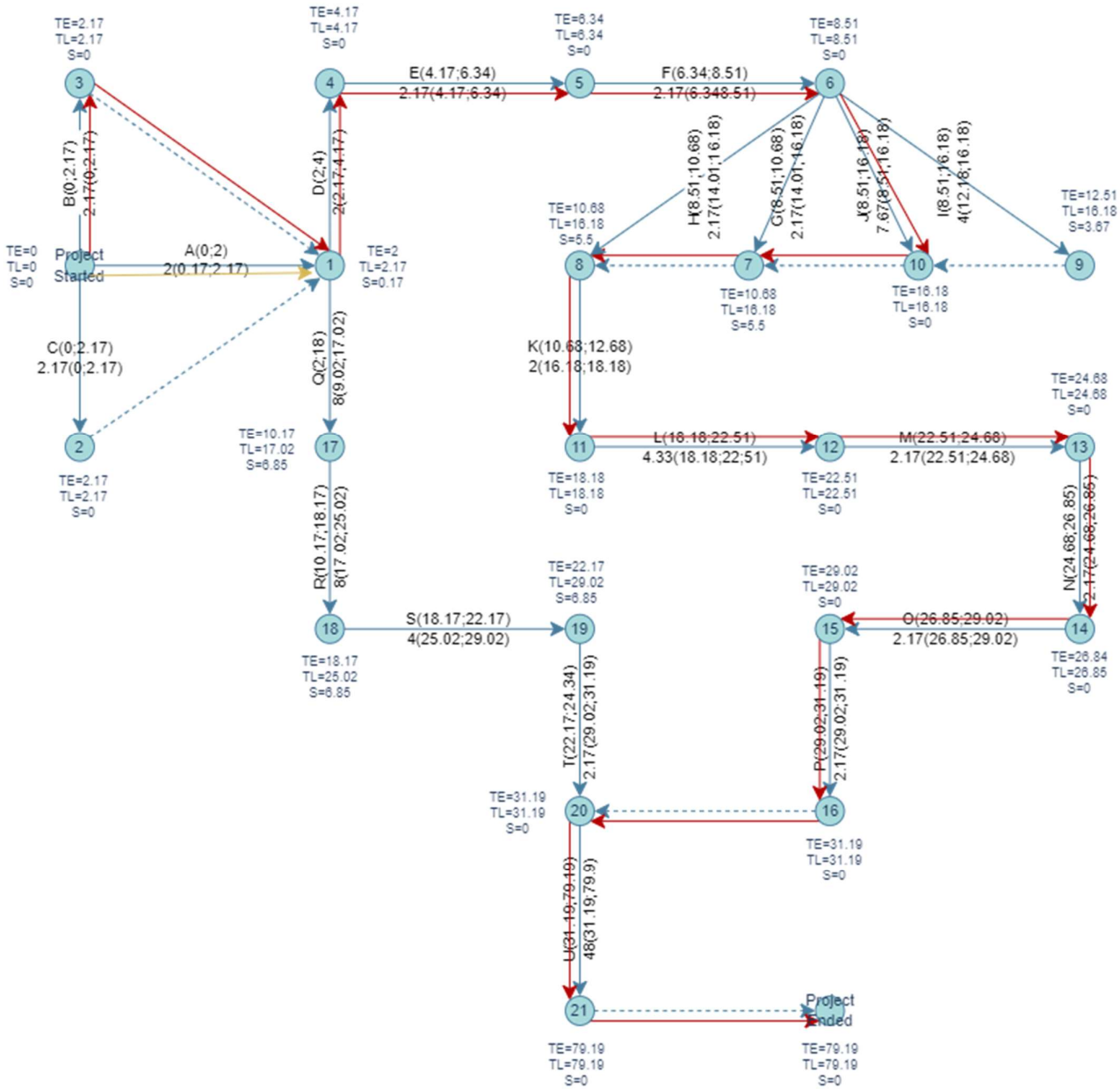

Figure 4: Activity times and floats for each activity with critical path on the network

Figure 5 Show the Activity times and floats for each activity with critical path which is highlighted with red arrow and subcritical path is highlighted with yellow arrow.

Conclusion:

This project encompasses a well-structured plan from pre-demolition to redevelopment, meticulously laid out in logical phases with corresponding budget allocations, personnel requirements, equipment needs, and time estimations. The detailed network diagram, Gantt chart, and critical path analysis provide a comprehensive overview of the project's timeline and potential bottlenecks. With thorough planning and efficient execution, the project stands poised for success, meeting its objectives within the projected time frame and budget constraints.